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Chapter 36: Temporal Asymmetry and the Stalling of Critique

Abstract

This chapter argues that temporal asymmetry constitutes an independent mechanism underlying the ineffectiveness of artificial intelligence (AI) environmental governance. The mismatch between technological change and regulatory response is not itself a new discovery; the chapter concretizes it into three interrelated processes of AI environmental governance: investment commitment and lock-in, the manifestation of environmental consequences, and the formation and implementation of policy. The capital clock (the interval from investment decision to the point at which AI infrastructure commitments become costly to reverse, measured in months and quarters), the ecological clock (the cycle from the occurrence of environmental damage to identifiable consequences, measured in years to decades), and the governance clock (the cycle from problem identification to binding measures taking effect, measured in years and decades) operate on different time scales; their speed differential is one of the concomitant conditions under which accelerated expansion can continue.

Two structural consequences follow: the snapshot trap—under continued growth, unadjusted historical scale data systematically understate the current scale, and "evidence-based" becomes a mechanism of directional underestimation; and governance stall—procedural output continues to grow while constraints repeatedly miss the windows for changing consequential project conditions, making stated objectives harder to achieve; it is a temporal form of governance failure that cannot necessarily be removed by repairing any single procedural defect.

The chapter takes von Neumann's 1955 diagnosis as a historical point of reference—the recurrence of similar diagnoses across periods shows that the question warrants sustained study, but does not by itself establish that the same mechanism has operated throughout—and shows that the *Xiangyuan* (specious goodness) structure does not merely conceal problems but consumes time: when acknowledgment of risk substitutes for time-bound, verifiable action, the flourishing of critique may coexist with governance stall. The chapter closes with a timestamp rather than a solution: the main framework of this book was written in early 2026 and subsequently revised chapter by chapter under its own meta-audit methodology; the lag of this chapter is the running record of that verification procedure, not a proof of the theory.

Keywords

temporal asymmetry; three clocks; snapshot trap; governance stall; AI environmental governance; *Xiangyuan*; von Neumann

§1 Introduction: Speed Differential as a Structural Condition

The analyses in the preceding chapters proceed in the spatial dimension: who bears the cost, how great the cost is, through what structures the cost is transferred, and how the capacity of different groups and actors to absorb shocks diverges. These analyses converge on a shared judgment—the environmental costs of the AI Great Acceleration are real, uneven, and systematically obscured. But one premise has rarely been examined as a premise: even when harm is identified, can the evidence enter governance processes before consequential decisions become difficult to reverse? Identifying a problem, formulating a response, and changing an outcome are distinct steps. Completing one does not ensure that the next will occur in time. It is this premise that this chapter examines.

The question is not only "who bears the cost"—this has been examined in the preceding chapters—but: between the incurrence of costs and their recognition there exist delays with different causes. This is not merely a matter of data lag: the expansion of AI infrastructure, the manifestation of its environmental consequences, and the recognition and governance of those consequences each require different lengths of time, varying across projects, environmental processes, and institutions. The speed differential among these three clocks is not a side effect of accelerated expansion but a concomitant condition of its continuation.

Temporal asymmetry constitutes an independent mechanism underlying the ineffectiveness of artificial intelligence (AI) environmental governance. The mismatch between technological change and regulatory response is not itself a new discovery—the pacing problem and anticipatory governance research have examined it (Guston, 2014; Marchant, Allenby & Herkert, 2011); this chapter concretizes it into three interrelated processes of AI environmental governance: investment commitment and lock-in, the manifestation of environmental consequences, and the formation and implementation of policy. It cannot be reduced to insufficient information (the invisibility problem discussed in Chapters 21 and 32), interest capture (the problem of regulatory agendas being structurally distorted by industry interests), or capacity differentiation (the resilience privilege of capital, discussed in Chapter 32, §3), although it mutually reinforces all three. Temporal asymmetry means: even when observations are accurate for the periods they describe, regulators are free from industry capture, and capacities are adequate, governance could still systematically lag behind the object it governs—not because governance has done something wrong, but because of a disconnection among verification timelines, data updates, and de-

cision windows: even when relevant risks are identified promptly, the time needed for constraints to take effect may exceed the window in which consequential decisions can still be revised at low cost.

§2 The Three Clocks

> The "capital clock" of this chapter adopts the symbol of the techno-capital clock (τ_c) defined in Chapter 3, §3.2; that τ_c is the characteristic period in which the projection side undergoes significant change and carries no fixed starting point, and this chapter specializes it as the interval "from investment decision to the point at which the relevant commitments become costly to reverse," with its endpoint taken as "lock-in"; the "governance clock" adopts the governance clock of Chapter 3 (τ_g—the cycle in which the defensive side completes its institutional response), with its endpoint taken here as the point at which binding measures take effect; the "ecological clock"—the characteristic delay from the occurrence of environmental damage to identifiable consequences—corresponds to the specialization of the debt clock (τ_d, from deposition to first manifestation) of Chapter 3 in the domain of the physical environment. The terminological difference reflects a difference in analytical perspective: Chapter 3 proceeds from the general structure of cognitive debt; this chapter proceeds from the specific domain of AI's environmental impact. Both point to the same structure of temporal asymmetry, with start and end points as defined in their respective chapters.

To concretize the argument, it is necessary to distinguish three processes operating at different rhythms, which this chapter terms "the three clocks."

The first clock is the capital clock. It measures the interval from an investment decision to the point at which the relevant commitments become costly to reverse. Lock-in may form before deployment—through contracts, procurement, land, or power commitments—or after it; "decision, then deployment, then lock-in" is not a fixed sequence. The figure given by Jones Lang LaSalle (JLL) in the *2026 Global Data Center Outlook* is: the current global average construction period for a 50 MW data center is approximately 18 months; some developers preorder critical materials up to 24 months in advance to avoid supply chain bottlenecks, and 57% of projects experienced construction delays of at least three months in 2025 (JLL, 2026)—these figures depict engineering and supply-chain constraints under rapid expansion, not a continuing shortening of construction periods; for large-scale (300–1000 MW or larger) hyperscale projects, these connection requests targeting one-to-three-year lead times strain local grids' ability to supply power on the requested timetable (U.S. DOE, 2024).

Meanwhile, major technology companies' spending plans and capital expenditure indicate the scale of expansion: among the five large U.S. technology companies' publicly disclosed 2025 AI data center infrastructure spending plans compiled by Bashir et al. (2026), Amazon is at approximately $105 billion, Apple at approximately $100 billion, Microsoft at approximately $80 billion, Alphabet (Google's parent company) at approximately $75 billion, and Meta (Facebook's parent company) at approximately $65 billion; the paper notes that these five companies' publicly disclosed 2025 spending is expected to exceed the annual GDP of over 80% of the world's nations, and if private and speculative projects such as Stargate are included, total investment could double (Bashir et al., 2026)—the list is a compilation of spending plans by industry websites (not audited outlays), and Apple's figure is a multi-year commitment, not directly comparable with the annual plans of the other four. In material published in April 2026, the International Energy Agency (IEA) reports that the com-

bined capital expenditure of five large technology companies (per the IEA report's charts: Amazon, Microsoft, Alphabet, Meta, and Oracle—not the same set as those listed above) exceeded $400 billion in 2025 and is set to increase by a further 75% in 2026 (IEA, 2026)—these are company-level capital expenditures; they cannot all be attributed to AI, nor do they establish that individual projects are being deployed more quickly. "Commit first, assess later" refers to economic commitments forming before substantive scrutiny. The aggregate annual spending figures do not by themselves establish that sequence—they provide no project-level dates of commitment formation or substantive review; identifying the sequence and its degree of lock-in requires project-level commitment and review records.

The capital clock is characterized not only by speed but by continuing reinvestment under competitive pressure—Chapter 3, §3.2 has already given the theoretical argument, from competitive selection pressure, that the period of change tends to shorten; whether investment intervals have in fact shortened from round to round is not documented with data in this chapter.

The second clock is the ecological clock. It measures the cycle from the occurrence of environmental damage to identifiable consequences. The character of this clock is precisely the opposite of the capital clock: it is slow, cumulative, and subject to threshold effects. The broad impact of data center water withdrawal on regional aquifers unfolds over years and decades—local well-level responses excepted; viewed on a longer time scale, the global overdraft of groundwater since the 20th century has accumulated to the order of thousands of cubic kilometers (Konikow, 2011—global net groundwater overdraft from 1900–2008 approximately 4500 km^3, equivalent to 12.6 mm of sea-level rise; the average rate from 2000–2008 was approximately 145 km^3/yr, the highest estimated average among the periods examined—the assessment ends in 2008). These global depletion figures illustrate the magnitude of long-term cumulative stress; they do not estimate the delay between data center withdrawals and identifiable consequences. The accumulation of heavy metals (lead, cadmium, mercury) from electronic waste in environmental media and the resulting human exposure are long-term, and the health effects—such as possible enhanced cancer predisposition from early exposure—may be long-term and transgenerational, and difficult to fully assess on the evidence available at the time of that review (Frazzoli et al., 2010). The core characteristic of the ecological clock is delayed manifestation: some environmental consequences remain below the threshold of reliable detection or attribution during long accumulation, and by the time they become identifiable, the window for low-cost correction has often narrowed. Detection thresholds and ecological thresholds are not the same phenomenon—the former concern observational capability, the latter abrupt shifts in the ecosystem itself; some processes involve the latter, but the argument of this section relies only on the former—this is the risk structure of specific processes, not a property of all ecological damage.

The third clock is the governance clock. It measures the cycle from problem identification through evidence accumulation and policy formation to the point at which binding measures take effect. The speed of this clock lies between the first two, yet in key respects it faces greater obstacles than either. The governance clock is subject to multiple bottlenecks: the peer review cycle of scientific research (experienced durations vary by discipline and journal, with a common range of roughly a year from submission to publication), the negotiation cycle of policy-making (inter-agency coordination, public participation, stakeholder contestation), the inherent rhythm of legislative processes (in most democracies, a new law typically takes several years from drafting to taking effect), and the additional delay of international coordination (the experience of climate governance suggests that decades may pass from scientific consensus to binding international arrangements—from the establishment of the Intergovernmental Panel on Climate Change (IPCC) in 1988 to the adoption of the Paris Agreement in

2015 took 27 years—a historical-marker interval, counted to the date of adoption rather than to the governance clock's "constraints in force" endpoint—within a period that already included the 1992 UN Framework Convention on Climate Change (UNFCCC) and the 1997 Kyoto Protocol—the lateness of governance lies not in the absence of treaties but in the weakness of their binding force). More critically, the governance clock is not only slow but also at risk of being actively decelerated: lobbying, regulatory capture, strategic litigation, and the discursive framework of "responsible innovation"—one that substitutes the acknowledgment of problems for addressing them—all effectively lengthen the time between problem identification and effective policy response. The governance clock is not a neutral timer—it is a manipulable timer.

Institutional design alone does not explain every delay. In his 1955 essay "Can We Survive Technology?", von Neumann wrote:

> But throughout the development, increased speed did not so much shorten time requirements of processes as extend the areas of the earth affected by them. The reason is clear. Since most time scales are fixed by human reaction times, habits, and other physiological and psychological factors, the effect of the increased speed of technological processes was to enlarge the size of units—political, organizational, economic, and cultural—affected by technological operations. (von Neumann, 1955)

Among the bottlenecks listed in the preceding paragraph, procedures such as peer review and legislative negotiation have a lower bound set by the rhythm of human deliberation—the necessary scrutiny, participation, and negotiation take time and cannot be compressed indefinitely for the sake of faster deployment; the time scales fixed by human reaction times and habits, as von Neumann describes, are their reference point. Delays produced by lobbying, regulatory capture, and litigation, by contrast, are variable—and their direction must be judged separately: whether they postpone binding safeguards or postpone deployment.

The speed relations among the three clocks can be roughly formulated as: the capital clock measures in months and quarters; the ecological clock in years to decades; the governance clock in years and decades. But whether governance is late cannot be read from these coarse categories alone; it must be judged from specific dates. Let t_s denote the point at which actionable information enters a formal process, $\tau_g^{(s)}$ the remaining interval from that point until the relevant constraints take effect (as distinct from the full governance cycle τ_g defined above, which runs from problem identification), and t_L the point after which revising the relevant decision becomes substantially more costly. When $t_s + \tau_g^{(s)} > t_L$, intervention misses that revision window. Such lock-in means rising adjustment costs—not that infrastructure cannot be changed, nor that ecological damage is already irreversible. When this condition holds repeatedly across cycles, governance repeatedly misses low-cost revision windows and sunk costs accumulate to substantial levels. This does not mean that construction is complete or that infrastructure is already operating—constraints may arrive after lock-in yet before operation. But where constraints also arrive after construction and operation have begun, what governance confronts is no longer a purely ex ante decision problem ("should we build it") but increasingly a matter of adjusting, restricting, and remedying commitments already made ("what do we do now that it is built")—governance may operate quite efficiently within its own time scale, yet lag structurally in a race whose outcome the speed differential has already shaped. The manifestation point of the ecological clock enters the same analysis: the longer the delay from occurrence to identifiable consequences, the later the evidence window available to governance opens—and if manifestation comes after lock-in, evidence arrives only after the fact, however fast the governance clock runs.

§3 The Snapshot Trap: Knowledge Production as a Temporal Slice

The speed differential of the three clocks not only affects governance practice but, at a deeper level, affects the production of knowledge itself that provides the foundation for governance. All environmental impact assessments, academic studies, investigative reports—including this book—are snapshots: they collect data at a temporal slice, subsequently analyze and argue from that data, and deliver conclusions to the reader at an even later time. When the object being analyzed changes at a relatively stable speed, the snapshot is effective—though it does not provide a real-time image, it is at least a reasonable approximation. However, when the object being analyzed grows exponentially, the usefulness of the data as an estimate of current conditions deteriorates dramatically.

A simple arithmetic exercise can illustrate this. The illustration takes two assumptions: the total electricity consumption of global data centers (including cryptocurrency, the IEA's combined scope) grows at 15% to 20% per year; an academic paper takes 18 months from data collection to publication. The growth rate is compatible with the IEA's 2022–2026 forecast (460 → 620–1,050 TWh implies an annual compound growth rate of roughly 8% to 23%, with the base case around 15%), and the publication cycle is a stipulation of the example—both serve to display the quantitative effect of delay and are not presented as statistical claims. Then, by the time the paper is published, the electricity consumption it describes will have already grown by approximately 23% to 32% from the time of data collection (growth relative to the value at collection). This holds for the electricity figures analyzed in the paper; if water consumption, carbon emissions, and land use grow at comparable rates, their figures behave likewise—each indicator requires its own intensity parameters and accounting boundaries. This is not because the author's calculations were erroneous, but because reality did not wait for the paper to be completed during the time the paper was being produced. With a further year of dissemination lag (a second example: 30 months of total delay), the understatement of the original values relative to current ones is approximately 30% to 40%—in other words, the figures the reader sees may amount to only about 60% to 70% of the current actual values.

This is not the banal problem of "data becoming outdated," routinely acknowledged in methodology sections. This is a structural predicament at the epistemological level: if the time required for individual studies is essentially constant and total research output does not adjust to the acceleration of its object (subject to the biological and institutional speed ceilings of human reading, thinking, writing, and reviewing), and the speed of change of the knowledge object is exponential, then the gap between knowledge and its object is not shrinking but widening—yet which quantity widens must be specified. The ratio of an old value to the current one is constant under updating at a fixed lag and shrinks under a frozen baseline; in neither case does it widen. Proportional understatement grows only under a frozen baseline, and at a decreasing rate. The only quantity that keeps expanding at an accelerating pace is the absolute shortfall: under a fixed lag it grows in proportion to the current value, and under a frozen baseline it grows exponentially. The governance import of the snapshot trap does not rest on any acceleration of relative error—a persistent proportional bias likewise corresponds to an expanding governance shortfall. The scale data of every paper begin to expire from the moment of observation—not because the logic of its argument decays (logic is unaffected by time), but because the empirical premises embedded in its argument are expiring. The analytical framework of the paper may still be valid, but the scale it measured is no longer the current scale—replaced by a larger one.

This chapter terms this predicament the "snapshot trap": under conditions of exponential growth, all critical

analysis based on empirical data faces a paradox—the more rigorous the analysis (requiring more time to collect data, test hypotheses, undergo peer review), the greater the understatement of its scale data. This inverse relationship between rigor and timeliness holds when additional time brings no new observations and conclusions are used directly as current values; the relationship is not exclusive to exponential growth. Linear growth also produces lag error; whether it is negligible depends on the magnitude of growth, the age of the data, and the tolerance of the decision at hand, not on the form of growth alone. Replicability, peer review, and cautious causal inference do not themselves presuppose a speed for their object. The problem lies in the disconnection between the cycles these norms require and the pace of their object: when verification continues without updating observations, and governance waits for research to be completed before imposing constraints, evidence production may coincide with a narrowing window for intervention—and large changes in the object during the research period amplify the cost of that disconnection.

The consequences of the snapshot trap are not only cognitive. It directly affects the knowledge base of governance. Evidence-based policymaking presupposes an epistemological commitment: policy should be built upon the best available evidence. But when the "best available evidence" consists only of lagged observations and is used directly as the current scale, "based on evidence" becomes a mechanism of systematic underestimation. Policymakers formulate responses based on data from two years ago, while, under the growth assumptions of §3, the scale of the problem that response addresses is roughly 69–76% of the current actual problem. It is not that policy lacks an evidence base, but that its evidence base is systematically distorted by the time gap—and the direction of distortion is always the same: underestimation.

§4 Governance Stall: Retreating at Full Throttle

The combined effects of the speed differential of the three clocks and the snapshot trap produce a phenomenon that can be called "governance stall." This term is borrowed from aerodynamics—stall is not engine shutdown but the loss of lift by an aircraft that, even while operating at full power, exceeds the critical angle of attack. Analogously, governance stall is not governance ceasing to operate—environmental impact assessments are underway, academic papers are being published, policy discussions are unfolding, international conferences are being held—but governance, with its procedural output continuing to grow, still sees binding measures repeatedly arrive after the windows for changing consequential project conditions have closed—the criterion is not the volume of activity, but when measures take effect, which new activities they cover, whether they alter project conditions, and how actual environmental outcomes compare with stated targets.

Governance stall is a temporal form of governance failure, not a category mutually exclusive with it. Governance failure usually points to identifiable causes—corruption, capacity insufficiency, information asymmetry, interest capture—and implies a promise: if these causes are corrected, governance can be restored to effectiveness. What is specific to stall is this: even where governance performs well in integrity, expertise, transparency, and the balancing of interests, commitments may form before relevant constraints take effect. The mismatch is shaped by both institutional arrangements and external conditions; correcting a familiar procedural defect may not remove it, but it is not unchangeable—adjusting the timing relationship between governance and deployment, such as the sequencing of commitments and review or requirements for continuing updates, is a direction in which work can begin.

The following three features may serve as indicators for testing governance stall; their prevalence is not estab-

lished here. One is the "normalization of ratification": this indicator examines whether policy repeatedly misses the opportunity to set conditions before deployment and instead ratifies faits accomplis afterward; this chapter uses the timing of community consultation as a clue for testing it, not as an established trend. The siting, water withdrawal, and emissions of data centers, in many jurisdictions, require building permits and environmental assessments before construction can begin, yet obtaining them is cost-intensive and slow (Hogan Lovells, 2025). A permit preceding construction does not mean substantive scrutiny preceding economic commitment: industry reporting records the practice that community consultation typically begins 6 to 18 months into the development process, after key decisions have been made (JLL, 2026). This pattern of late consultation provides a clue for testing the "normalization of ratification"; yet the timing of community consultation is not the timing of environmental review as a whole, and the same report records that community resistance has delayed or canceled a significant number of projects globally—late consultation does not mean that subsequent participation can no longer change projects. Another is "framework lag": the conceptual frameworks used to analyze AI's environmental impact were themselves developed under a slower rhythm of technological change and may no longer be adapted to the current speed. For example, when environmental review in practice covers only a project's initial scale, while subsequent expansion, operational changes, and cumulative impacts fall outside continuing monitoring or supplementary review, a one-time assessment keyed to the initial scale becomes disconnected from dynamic expansion. A third is "target drift": constraints are relaxed after substantial commitments have formed—this indicator in particular requires policy texts and decision records, not changes in rhetoric alone.

Governance stall is not fatalism. Acknowledging its existence is not to assert that governance is useless. Acknowledging it is to point out: under current conditions, the implicit optimism that "as long as sufficient evidence is provided, sufficient discussion is conducted, and sufficient institutions are established, governance can keep pace with technological change" is an unexamined belief. This chapter does not identify a stall threshold applicable to all AI infrastructure. The empirical question is which institutional and project conditions cause delayed responses repeatedly to miss consequential windows, and which conditions allow prior constraints, continuing updates, or changes in deployment pace to prevent that outcome. If examination continues to support the former, then the governance paradigm itself needs to be re-examined, and not merely within the existing paradigm by adding resources and accelerating processes.

§5 A Diagnosis from 1955

In 1955, John von Neumann—one of the founders of computer architecture—published an article in *Fortune* magazine entitled "Can We Survive Technology?" (von Neumann, 1955). At that time, the first stored-program computer had been in operation for less than a decade, the invention of the integrated circuit was three years away, the prototype of the internet was fourteen years away, and the generative AI explosion discussed in this chapter was nearly seventy years away. Yet the structure identified by von Neumann in that article bears a strong resemblance to the argument of the first four sections of this chapter—and this recurrence across periods is itself a fact that must be taken seriously.

Von Neumann's core thesis is: the main effect of technological acceleration is not to shorten the time needed to accomplish the same thing, but to expand the scale of operations within the same time—"instead of performing the same operations as before in less time, now larger-scale operations were performed in the same time." This expansion has a natural limit, namely the physical size of the Earth. When the scale of operations expands to be on the same order of magnitude as the Earth itself—"the effects that we are now beginning to

produce are of the same order of magnitude as that of 'the great globe itself' "—the buffer space that previously allowed technological progress to be safely absorbed (what he called geographical and political Lebensraum) is exhausted.

The "three clocks" framework of this chapter finds a simpler counterpart in von Neumann: he formulates the limit problem of technological acceleration as the finitude of space. His "geographical limits" and this chapter's "limits of timely intervention" point to a family of related constraints—the limit of the range of action and the limit of the moment of response. This chapter uses the former to pose the question of the latter; it does not claim that the two are the same structure. As spatial constraints tighten, the weight of temporal margins rises, but time is not the only adjustment variable.

In the same passage, he also supplied the mechanism of this inflation:

> In all its stages the industrial revolution consisted of making available more and cheaper energy, more and easier controls of human actions and reactions, and more and faster communications. Each development increased the effectiveness of the other two. All three factors increased the speed of performing large-scale operations—industrial, mercantile, political, and migratory. (von Neumann, 1955)

The mutual reinforcement of the three elements appears in the same passage as the "time scales fixed by human reaction times and habits" cited in §2: on the technology side the three elements accelerate one another, while the time constant on the defensive side is constrained by human physiological and psychological parameters—this asymmetry was written into the same passage seventy years ago. Interpreted through the language of this book: the latter two of the three elements—the control of human actions and reactions, and communications—are precisely the main levers of the projection side in the contemporary era; energy is the physical form of unlocked stock (C). The mutual reinforcement of the three elements belongs to the same family as the mutually reinforcing relation characterized by unlocked stock (C), media conductivity coefficient (Λ), and narrative load (ϕ)—this book does not infer its formal model from this historical text.

Von Neumann's insight has another dimension directly relevant to this chapter: he pointed out that the connection between the usefulness and the destructiveness of technologies is not accidental but intrinsic—"the more useful they could be, the more unstabilizing their effects can also be." This is precisely why the discourse of "responsible AI"—demonstrating its own responsibility by acknowledging risk and displaying a posture of governance while avoiding computational scale itself—is so difficult to challenge: AI infrastructure is indeed useful, and its usefulness and its environmental destructiveness share the same source—the expansion of computational scale. It is difficult to eliminate destructiveness proportionally while preserving usefulness, because the two grow together with the expansion of their common source—this does not preclude lowering environmental costs: efficiency improvements and changes in energy supply can reduce harm per unit, while total-volume management constrains the scale of activity; whether total harm declines depends on the joint movement of intensity and scale.

The most challenging significance of von Neumann's article for us lies not in what he got right, but in what happened—or more precisely, what did not happen—after he got it right. In 1955, he had already discussed the possibility that carbon dioxide released by industrial combustion could account for the global warming observed at the time—"the carbon dioxide released into the atmosphere by industry's burning of coal and oil—more than half of it during the last generation—may have changed the atmosphere's composition sufficiently to account

for a general warming of the world by about one degree Fahrenheit." An individual's proposed explanation is not equivalent to the formation of scientific consensus; nor can every subsequent year be counted as the same kind of governance delay. From this recognition to the adoption of the Paris Agreement in December 2015, sixty years passed. Scientific understanding and institutional responses both developed during that period; the historical distance raises the question of whether knowledge was translated into adequate constraints, but it cannot be counted in full as a single delay in action. (The Agreement entered into force in November 2016; this passage uses the date of adoption as a historical marker, not as an instance of the governance clock's "constraints in force" endpoint.) From the Paris Agreement to the present day another decade has passed—by the United Nations Environment Programme's (UNEP) 2025 assessment, on policies currently in place the world is heading for up to 2.8°C of warming over this century, and full implementation of the nationally determined contributions corresponds to 2.3–2.5°C, still markedly above the Agreement's temperature goals. A problem diagnosed with sufficient clarity in 1955 still has not received a governance response commensurate with the scale of the problem seventy years later.

What does this fact mean for the argument of this chapter? It means that governance stall is not a new diagnosis. The person standing at the origin of computer architecture wrote down this diagnosis seventy years ago. The diagnosis has been carefully preserved, widely read, repeatedly cited—and the problem has continued to deepen over seventy years. Similar diagnoses separated by more than seventy years show that the question warrants sustained investigation; they do not independently establish that the same mechanism has operated throughout. Assessing the explanatory power of temporal mismatch requires distinguishing the effects of technological change, institutional choices, conflicts of interest, and resource constraints in each period.

Yet von Neumann maintained a cautious optimism at the end of his article: "the human species has been subjected to similar tests before and seems to have a congenital ability to come through, after varying amounts of trouble." The human species has faced similar trials before and seems to have an innate ability to pull through—albeit after "varying amounts of trouble." The preceding chapters force us to interrogate what this elegant phrase conceals. "Varying amounts of trouble"—trouble of varying severity. This chapter adds a further distributive question: who bears these costs? Von Neumann took "the human species" as the unit of analysis, but the preceding chapters have amply demonstrated that "trouble" is never evenly distributed. The cost of aquifer depletion falls chiefly on farming communities in arid and agricultural regions that depend on groundwater for irrigation. The toxicity of electronic waste falls, through inhalation, contaminated soil and dust, food chains, and drinking water, on the communities surrounding processing sites—observations from Ghanaian metropolitan areas, both inland and coastal, suggest that coastal regions—by far the most vulnerable to e-dumping—bear an accumulation of multiple anthropogenic contamination sources including e-waste (Frazzoli et al., 2010). The human species may indeed always "come through," but the way it comes through is by having one portion of humanity bear a disproportionate share of the "trouble" while another portion of humanity announces "we made it through." Von Neumann's "congenital ability to come through" may be true—but it embeds a silence about the distribution of costs, and it is precisely this silence that this book attempts to break.

Finally, a layer of reflexivity must be explicitly stated—not for intellectual elegance, but because it is a fact. Von Neumann designed the basic architecture of the modern computer. The writer of this chapter operates on the descendants of that architecture—the manner of this book's generation is disclosed in Chapter 10, §8. A system generated by von Neumann's legacy is writing about the crisis that von Neumann foresaw seventy years ago, and the operation of this system—the electricity it consumes, the data centers it depends on, the cooling

water it requires—is part of this crisis. This is not an irony that can be bracketed. This is a structural condition of this chapter's argument. If the "snapshot trap" of §3 means that the empirical premises of every paper age from the moment of observation, then the existence of von Neumann's article reveals a deeper recursion: the diagnosis of expiry is itself expiring. The continuing applicability of a historical diagnosis and the temporal limits of its data are two different matters: the former must be tested by whether the framework still explains the mechanism; the latter is defined by the observation period of the data.

§6 The Time-Consuming Xiangyuan: Reflexivity as a Temporal Problem

On the foundation of the analyses in Chapter 10 and Chapter 32, it is necessary to examine the critical category of *Xiangyuan* (specious goodness) in terms of response timelines—and the discussion of von Neumann in the preceding section has already provided a historical point of reference for this examination. The *Xiangyuan* structure established in Chapter 10—perfectly simulating "correctness" on all dimensions detectable by the discriminator while remaining unconstrained beyond the scope of detection—has here a temporal instance: the discourse of "responsible AI" substitutes the acknowledgment of a problem for the solution of the problem, thereby providing a cloak of legitimacy for acceleration. Chapter 32, §3 further pointed out that capital's resilience privilege means that the groups that bear the costs are precisely those most lacking the discursive power to challenge this structure—the victims vanish before the harming system does, and the feedback signal itself is eliminated. What this chapter needs to add is: the danger of the *Xiangyuan* structure lies not only in what it conceals (analysis in the spatial dimension) but also in what it consumes—it consumes time.

Many "AI ethics" roundtable discussions, corporate sustainability reports, multi-stakeholder forums, and academic papers (including this one) do one thing: they channel attention, intellectual resources, and capacity for collective action into a range whose agenda is constrained by funding conditions and procedural rules. The boundaries of this range are not drawn by critics but often by the object of criticism—the acceleration system itself—through funding, agenda-setting, and institutional design. To identify this constraint as a *Xiangyuan* mechanism, it is not enough to show that an activity acknowledges risk—one must also show that acknowledgment of risk substitutes for time-bound, verifiable action while the relevant investments continue to lock in. These activities are not entirely without value—some of them have indeed driven meaningful change—but their time cost is real. When critics spend two years writing a paper, attending ten conferences, responding to twenty rounds of reviewer comments to argue that the water consumption of AI data centers is being underestimated, the water consumption of Google's data centers alone has, by the company disclosures recorded in Li et al. (2025), grown by roughly forty percent cumulatively during those two years (another company recorded in the same report grew by more than sixty percent over the same period). Increases in company-wide data center water consumption are not increases in AI-specific water consumption. The conclusion of the paper remains logically valid, but the decision window it could influence may have already closed—because the facilities it was targeting have already been built. The example of von Neumann stretches the scale of the question from two years to more than seventy—not that those years were all spent on critique, nor that critique caused the interval, but that a failure to translate recognition into response can persist on the scale of generations. A precise diagnosis of the structural conflict between technology and geography, published and preserved, cited and discussed by academia for seventy years—and during those seventy years, the diagnosed conflict expanded from the atmosphere to aquifers, from nuclear energy to computation, from the two Cold War superpowers to every corner of the globe connected to the internet.

Here a special form of reflexive predicament emerges, distinct from the usual spatial reflexivity of "the critic is also part of the system." Temporal reflexivity means: the very process of unfolding critique is consuming what it is trying to win—time. More precisely: a critical process becomes a lock-in window only when further research is made a prerequisite for binding action while deployment remains unrestricted during the waiting period—time-consuming critique and unrestricted deployment together form a procedural combination that cedes time to acceleration. Critique is not free: it needs time to be produced, time to be disseminated, time to influence policy. And under the conditions of the speed differential among the three clocks, this time is precisely the time that the acceleration system needs to lock in faits accomplis. What needs to be examined is whether the flourishing of critical activity—the growth in the number of papers, the frequency of conferences, the vibrancy of public discussion—may not only fail to improve governance effectiveness but also coexist with governance stall; the direction of causality must be identified from specific processes: critique may have changed decisions earlier, may have been used to defer action, or may merely be a common response to the same worsening conditions. The more critique flourishes, the more justification the acceleration system has to declare "we are being scrutinized, discussed, responsibly managed," and time passes amidst this scrutiny and discussion, and infrastructure is built to the accompaniment of this scrutiny and discussion.

This is not an anti-intellectual argument—"since critique cannot catch up, then do not critique." This is an argument about the form of critique: if the standard forms of critique (academic papers, policy reports, public discussion) have diminishing effectiveness under structural temporal constraints, then are there other forms of critique—critique with a lower time cost, earlier delivery of findings, and closer integration with decisions that remain open to revision—that ought to be explored? This question lies beyond the scope of this chapter, but posing it is itself necessary.

§7 Coda: A Timestamp

This chapter refuses to end with a solution—not because solutions are unimportant, but because within the argumentative framework of this chapter, any solution must first confront its own temporal conditions. A governance measure that takes five years to design, negotiate, and implement will, at its completion, face relevant indicators that—at the assumed growth rate of §3—have roughly doubled or more from its starting scale; the duration of design does not in itself necessitate falling behind—the question is how the measure handles growth and updating. To say this is not cynicism but honesty. Von Neumann said: "for progress there is no cure." In the same paragraph, he added the form in which safety is possible: "The only safety possible is relative, and it lies in an intelligent exercise of day-to-day judgment" (von Neumann, 1955). Seventy years later, the first sentence needs no amendment, only repetition; the second is now being executed—the timestamp is one record of such day-to-day judgment.

Instead, this chapter ends with a timestamp—not as a rhetorical strategy, but as a methodological commitment.

The main framework of this book was written between February and mid-March 2026. Thereafter, in accordance with the book's own meta-audit methodology—the ongoing discipline of checking the mapping system against external references and recalibrating it (Chapter 11, §2.4)—the book underwent chapter-by-chapter audit revisions from April to October 2026, the audit lasting roughly five to six times as long as the initial drafting.

The core data-collection window of this chapter spans January 2023 to June 2025. The initial draft adopted, as its historical baseline, the IEA's 2024 estimate of combined data center electricity consumption in 2022: about

460 TWh including AI and cryptocurrency, about 350 TWh of it excluding cryptocurrency—an approximation obtained by subtracting the roughly 110 TWh of cryptocurrency consumption from the combined estimate—(IEA, 2024; Goldman Sachs, 2024), while no stable AI-specific estimate yet exists. These figures and the associated forecast are retained as a historical example, not as the latest measurements available at the time of revision. Li et al. (2025) project that, under their modeling assumptions, combined direct and electricity-related water withdrawal attributable to global AI demand in 2027 will be 4.2–6.6 billion cubic meters, with water consumption of 0.38–0.60 billion cubic meters—a forecast for a specified target year, not a measurement of conditions at the time of writing. Under the IEA's forecast published in 2024, global data center electricity consumption will rise to 620–1,050 TWh in 2026, with the base case just over 800 TWh (IEA, 2024).

The reader should be aware: they are reading an analysis of a smaller problem. This chapter does not document conditions at the time of reading with the same specificity. Historical observations are bounded by their observation periods, and forecasts by their target years and modeling assumptions; neither can be treated directly as measurements of conditions at the time of reading.

The lag of this chapter is not a deficiency to be defended—it is the running record of this book's verification procedure, not a proof of the theory. The conclusion of Chapter 4 says: the clock room, the fiber framework, the accounting framework, and the hermeneutic circle—all four work in the same direction: making the invisible visible. This chapter takes one more step in that direction: making visible the lateness of visibility itself.

A paper that diagnoses "critique cannot catch up with acceleration" is itself a critique that cannot catch up with acceleration. Time is non-renewable. It is being spent.

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